

Residual Stress Control Based on Electroplated Metal Grain Characteristics in Panel-level Glass Advanced Packaging

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Abstract— The low loss, coefficient of thermal expansion (CTE) match, and dimensional stability of glass in large panel manufacturing makes glass advanced packaging technology a great choice for chiplet systems. However, the residual stress of the metal layer generated during the TGV (through-glass-via) electroplating process is one of the major technical bottlenecks of glass packaging, resulting in glass-metal layer delamination and glass breakage. This paper demonstrated for the first time a method to achieve regulation of the residual stress by controlling the grain growth of the plated Cu in the electroplating process. Detailed relationships among the residual stress, Cu grain characteristics, and process parameters were studied during both the plating and annealing processes. During the electroplating process, test vehicles with different plating parameters (current density, accelerator, inhibitor and leveler ratio) were fabricated, then the Cu grain characteristics were characterized using focused ion beam (FIB), and the related orientation and residual stress were measured using the X-ray diffraction (XRD) technology. The results showed that there are four types of textures in the Cu plating layer, namely (111), (200), (220), and (311), and the residual stress of Cu film varied with the plating parameters changed. During plating, the residual stress and (111) texture showed a positive correlation, and nanotwins can help relieve high residual stresses. Additionally, the recrystallization of the plating grains occurred during the annealing process, accompanied by an increase in grain size while the residual stresses were reduced. Cu films with highly (220)-oriented or (111)-oriented were more stable during annealing and exhibited lower residual stresses after annealing. Ultimately, the residual stress in electroplated Cu was successfully regulated by optimizing the plating and annealing process.

Keywords—chiplet, glass packaging, TGV, nanotwinned Cu, residual stress

I. INTRODUCTION

The increasing demands for performance, functionality and heterogeneous integration ability of chiplet systems drive the evolution of advanced packaging technologies. Due to its excellent characteristics, glass substrates have great market potential to replace or to eliminate silicon interposers to achieve high-density, high-bandwidth, and low-power 2.5D glass advanced packaging systems, as shown in Fig. 1. The dimensional stability of glass in large panel manufacturing

makes glass packages a great choice for chiplet systems, which require large-area packaging for high digital applications. Therefore, the glass packaging is an ideal platform for heterogeneous integration of chiplet technology, and can effectively improve the performance of integrated circuits. However, the residual stress of the metal layer generated during the TGV (through-glass-via) electroplating process is one of the major technical bottlenecks of glass packaging, resulting in glass-metal layer delamination and glass breakage, which seriously restricts the mass production and large glass panel manufacturability.

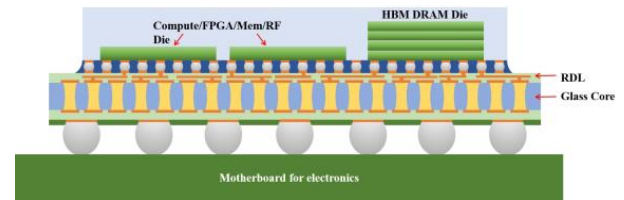


Fig. 1. Schematic diagram of glass substrate packaging system structure.

The present study on the failure analysis of TGV mainly focuses on the thermal stresses in TGV at the macro scale through finite element simulation and reliability testing methods. For example, Benali et al. developed a mathematical model of TGV with different geometries using finite element simulation to investigate the thermo-mechanical stresses in TGV [1]; Demir et al. investigated the stress-strain distribution in TGV by finite element simulations and thermal cycling reliability tests [2]. However, the current prior-arts lack the study on the residual stresses caused by changes in the characteristics of the plated metal grains at the microscopic scale. Though the research of the TGV microscopic failure mechanism is limited, it is still helpful to learn from other material's research methods on the intrinsic relationship between grain characteristics and residual stress. For example, Okoro and Wei et al. showed that microstructural changes of Cu grains during TSV electroplating could generate stress [3, 4]; Li et al. found that changes in current density and additives caused changes in the residual stresses inside Cu films [5]; Chen et al. claimed that the temperature of the plating solution affected the residual stresses in Cu films, and that the residual stresses in highly (111)-oriented nanotwinned Cu were higher than those in normal coarsened Cu [6].

In this study, we aimed to control the grain characteristics of plated Cu films by changing the plating and annealing

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process as a way to regulate the residual stresses of Cu inside TGV.

II. EXPERIMENT

A. Electrolytic Plating

The Cu layer was plated on a glass substrate with a 100 nm Ti adhesion layer and a 300 nm Cu seed layer. Plating solution (120 ml) contained 10-100 g/L H₂SO₄, 100 g CuSO₄, 20-60 ppm NaCl, and 0-300 ppm organic additives. Three additives were used in this study, namely poly-2-sulfur-2-propane sodium sulfonate (SPS), polyethylene glycol (PEG) and Tetronic (TEC). The direct-current (DC) plating experiments were done at an electrochemical workstation (Versa STAT 3A) with a current density of 2-12 A/dm² (amperes per square decimeter, ASD). Furthermore, the mechanical agitation rate was set to 1000 rpm to achieve uniform electrodeposition.

B. Annealing

After the plating work was completed, the prepared Cu film samples were heated inside a vacuum annealing furnace. The heating rate of the vacuum annealing furnace was 5 °C/min, and the set temperature of the furnace was 300 °C, holding time was 1.5 h. Then, the samples were naturally cooled to room temperature in the furnace. In order to prevent oxidation, the samples were kept in a vacuum environment during heating and cooling.

C. Characterization

Cross-sectional microstructures of plated Cu films were observed with a dual-beam focused ion beam (DB-FIB, Helios 5 CX), where the operating voltage was 30 kV and the operating current was 80 pA. X-ray diffraction (XRD, Smart Lab SE) was used to identify the texture and residual stress of the Cu films. The X-rays were scanned at a rate of 5°/min with a step size of 0.02°, and the scanning angle range was 30°-100°. The ratio of the different texture types was calculated according to Eq. (1):

$$R_{(hkl)} = \frac{I_{i(hkl)}}{\sum_{i=1}^n I_{i(hkl)}} \quad (1)$$

where $R_{(hkl)}$ represents the ratio of (hkl) texture in the whole plated Cu film, $I_{i(hkl)}$ represents the intensity of the (hkl) texture peak in the XRD pattern, and n is the number of texture types in the sample.

For residual stress measurements, each sample was tilted at four different Ψ angles, 0°, 15°, 30°, 45°, and a high angle (311) peak was chosen to measure strain. Specifically, the residual stress of Cu films was calculated by Eq. (2) [6]:

$$\sigma_\phi = -\frac{E}{2(1+\nu)} \cot \theta_0 \frac{\pi}{180} \frac{\partial(2\theta)}{\partial(\sin^2 \Psi)} \quad (2)$$

where E and ν were the Young's modulus and Poisson's ratio of the material, respectively, θ is the angle of the X-ray beam, and σ_ϕ was the residual stress of Cu films.

III. RESULTS AND DISCUSSION

A. Texture and residual stress in plated Cu films

Fig. 2 demonstrated the effect of current density on the grain texture of electroplated Cu films. For this set, the plating solution did not contain any organic additives. The XRD patterns showed that the prepared Cu films had four types of

textures, i.e., (111), (200), (220), and (311). However, the intensities of (200) and (311) textures were very low. As the current density changed (2 ASD → 12 ASD), the intensity of the (111) texture increased and then decreased, while the (220) texture showed exactly the opposite trend. At a current density of 12 ASD, the Cu film was almost entirely (220)-oriented grains. However, (111)-oriented grains dominated the growth of Cu films at 8 ASD. It can be seen that there was a strong relationship between the orientation of Cu film and current density.

According to the calculation of Eq. 2, the residual stresses of Cu films at four current densities (2, 5, 8, and 12 ASD) were 210.2, 232.2, 279.5, and 190.7 MPa, respectively. It is very interesting that the trends of residual stress and (111) texture content were consistent when the current density was increased. Therefore, it is speculated that the increase in (111)-oriented grains during plating without additives caused an increase in residual stresses.

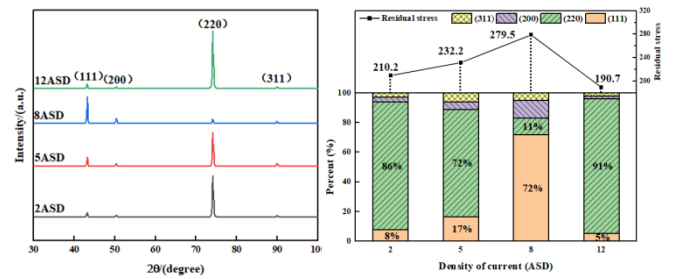


Fig. 2. XRD and residual stress results for different current densities.

Next, to investigate the effect of organic additives on residual stresses, we chose a current density of 8 ASD. Fig. 3 showed the XRD and residual stress results for different PEG concentrations. The addition of PEG resulted in a reduction of the (111) texture. Although the change in PEG content did not change the texture of Cu films, the residual stresses in the films were lower at higher PEG content (≥ 50 ppm).

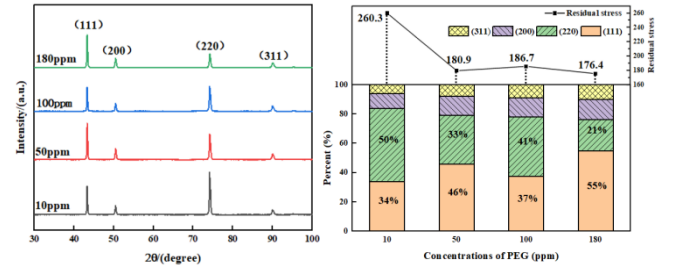


Fig. 3. XRD and residual stress results for different PEG concentrations.

Fig. 4 displayed the XRD and residual stress results for different SPS concentrations. For the preferred orientation, the results of SPS and PEG were similar. It is worth noting that the presence of these two additives resulted in an increase in the strength of the (200) and (311) texture. The values of residual stresses in Cu films with different SPS contents were all around 240 MPa. The residual stress of the film was reduced by approximately 14.4% compared to the case without additives. However, as the SPS content increased, the residual stresses in the Cu films gradually became larger.

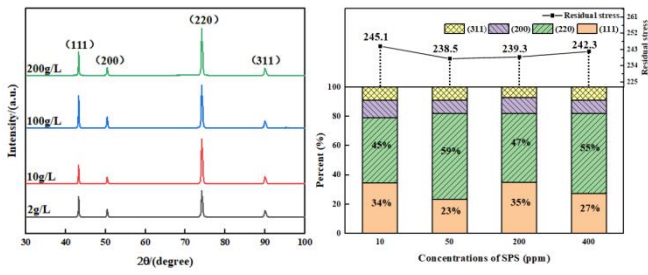


Fig. 4. XRD and residual stress results for different SPS concentrations.

Fig. 5 displayed the XRD and residual stress results for different TEC concentrations. The experimental results showed that the addition of a small amount of TEC did not change the texture of Cu film, but greatly reduced the residual stress. Compared to the case without additives, the residual stress was reduced by 25.1%.

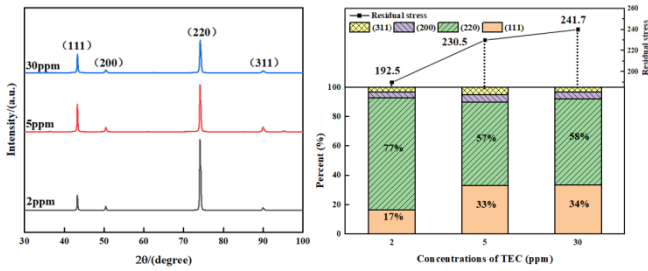


Fig. 5. XRD and residual stress results for different TEC concentrations.

Relevant literature demonstrated that the introduction of additives could change the overpotential, and different overpotentials induce different strain energies, which in turn affects the optimal growth of the grains [7, 8]. Chan and Kremmer et al. showed that high overpotentials were favorable for the formation of (220) texture due to the fact that the two-dimensional morphology of (220) texture can be minimized at high overpotentials [9, 10]. Therefore, we speculated that the addition of additives may lead to changes in the electrochemical characteristics of the plating solution, which in turn affected the grain characteristics and the residual stresses of the plated Cu films.

B. Effect of annealing on the texture and stress of Cu films

The experimental results described above showed that there seemed to be an intrinsic link between residual stresses and texture in Cu films. In order to verify this speculation, we prepared eight samples with different texture ratios by controlling the current density, and the type and concentration of additives in the plating solution. Fig. 6(a) illustrated the XRD results for samples with serial numbers 1-8. The percentages of (111) texture for samples 1-8 were 5.1%, 16.3%, 22.4%, 37.8%, 47.1%, 54.6%, 72.7%, and 94.4%, respectively, as shown in Fig. 7. In other words, there was a gradual decrease in the ratio of (220) texture.

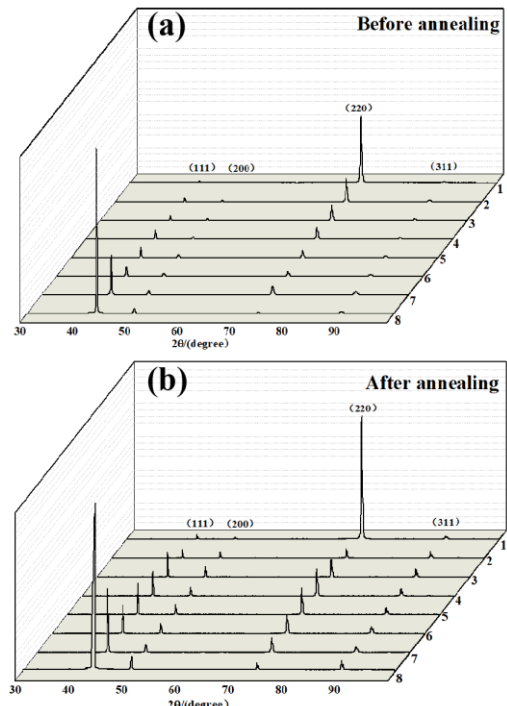


Fig. 6. XRD results before and after annealing of Cu films with different (111) texture ratios. (a) Before annealing. (b) After annealing.

Subsequently, these samples were subjected to an annealing treatment process at 300 °C for 1.5 h. The XRD results of the samples after annealing were given in Fig. 6(b). The results showed that samples corresponding to numbers 2-7 had changed in texture distribution, as shown in Fig. 7. Specifically, the annealing treatment resulted in an increase in (111) texture and a decrease in (220) texture. In addition, the (200) texture increased after annealing. Fig. 8 shows the FIB images of the sample cross section before and after annealing. It can be seen that these samples underwent recrystallisation process during the annealing, where the original irregular small grains were replaced by large grains and the grain size became larger.

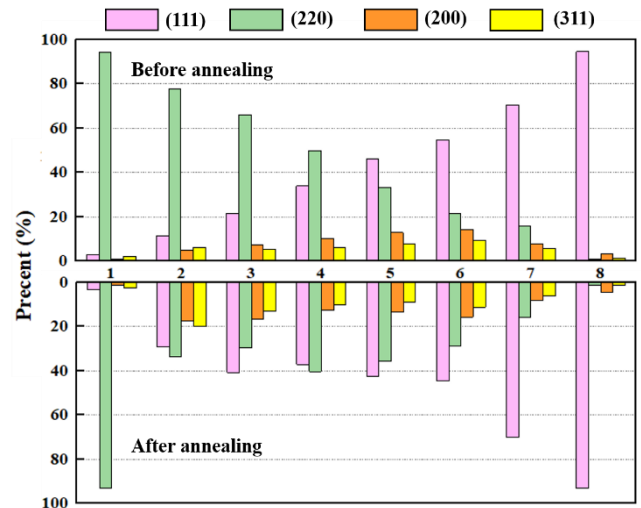


Fig. 7. Ratio of different textures of Cu films before and after annealing.

Chen et al. claimed that annealing caused the formation of (200) texture in Cu films [11]. However, in our experiments, a surprising phenomenon was that the texture of highly (111)-oriented and highly (220)-oriented (number 1 and 8) Cu films hardly changed. In other words, the (200)

texture did not develop. According to the FIB results (Fig. 8(a)), for the highly (220)-oriented Cu films, although the texture did not change after annealing, the grains underwent a full recrystallisation process and the grain size became very large. Compared to highly (220)-oriented Cu films, the phenomenon was totally different for the highly (111)-

oriented Cu film. According to Fig. 8, it can be seen that the microstructure of the highly (111)-oriented Cu films hardly changed after annealing, which may be due to the presence of a large number of lower grain boundary energy nanotwins inside the highly (111)-oriented Cu films (Fig. 8(h)) [11].

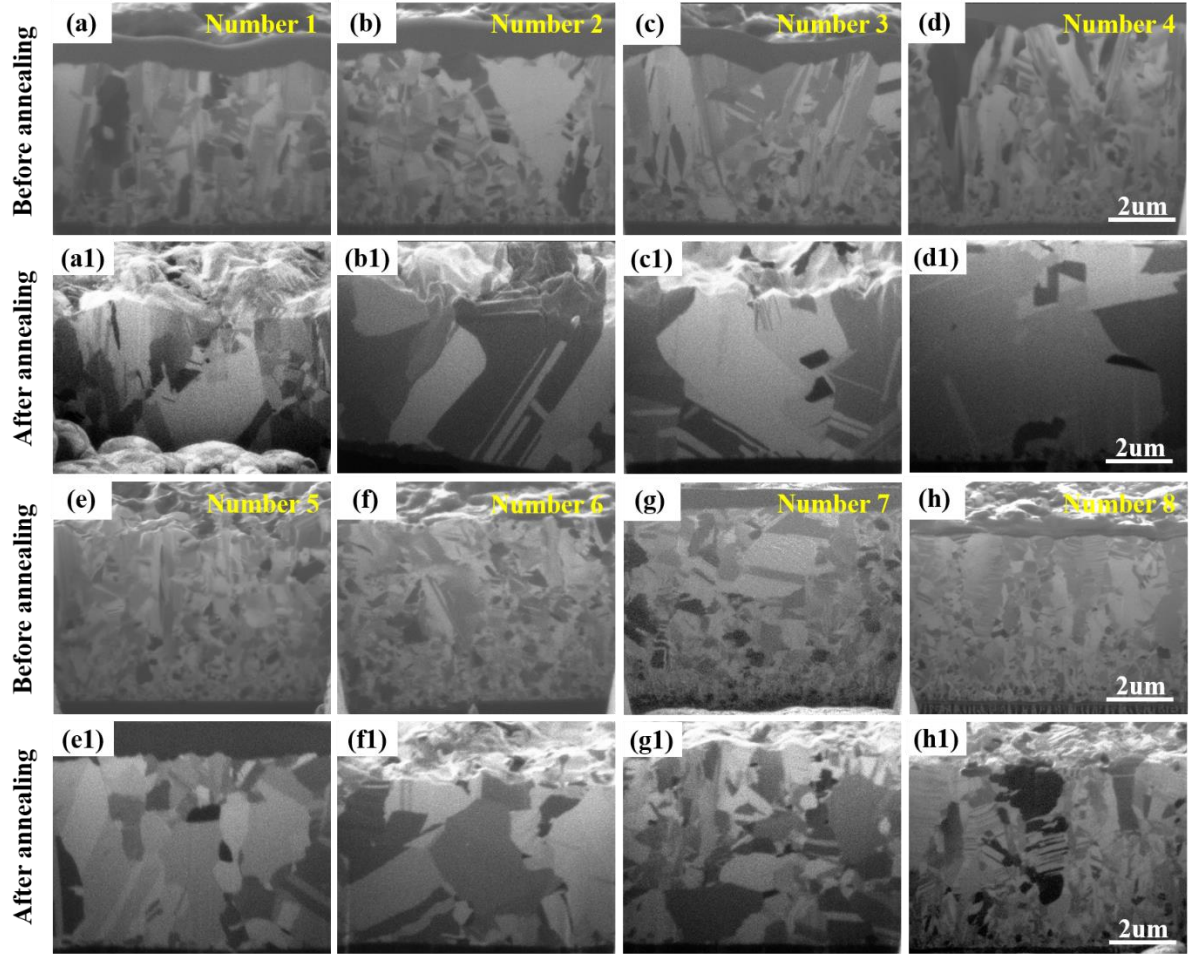


Fig. 8. FIB images before and after annealing of Cu films with different (111) texture ratios. (a-h) Before annealing. (a1-h1) After annealing.

Fig. 9(a) showed the residual stress of Cu films with different (111) texture ratios before annealing. The residual stress of the Cu film increased gradually with the increase of the ratio of (111) texture, implying that there was a strong intrinsic link between the residual stress and the texture. However, the rate of increase of residual stress decreased as the ratio of (111) texture increased. This may be due to the fact that the proportion of nanotwins in the Cu films increased as the (111) texture increased, because the formation of nanotwins resisted a portion of the residual stress [6, 9].

Fig. 9(b) showed the residual stress after annealing of Cu films with different (111) texture ratios. The results showed that the annealing process drastically reduced the residual stress in the Cu films. The residual stress was below 62 MPa for all samples. In addition, after the annealing treatment, the residual stresses were minimized for the highly (111)-oriented Cu films, followed by highly (220)-oriented, which suggests that the highly (220) and (111) texture can greatly relieve the residual stresses of Cu films.

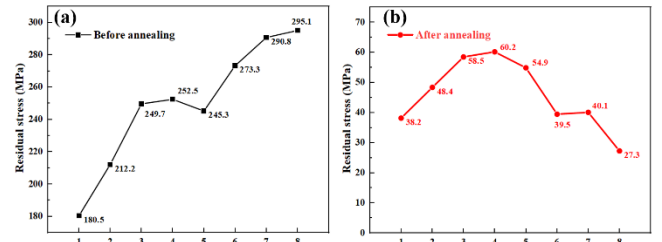


Fig. 9. Residual stress before and after annealing of Cu films with different (111) texture ratios. (a) Before annealing. (b) After annealing.

IV. CONCLUSION

This paper demonstrated for the first time a method to regulate the residual stress of TGV plating, and achieved the purpose of microscopically regulating the residual stress of TGV plating through the specific study of the complex change of the grain characteristics and residual stress during the electroplating and annealing process. The main results were as follows:

1) The two dominant orientations in electroplated Cu films were (111) and (220). With the increase of current density, both the (111) texture ratio and residual stress increased first

and then decreased. Organic additives can change the texture and residual stress of Cu films;

2) With the increase of (111) texture ratio, the residual stress increased gradually. The nanotwins in highly (111)-oriented Cu films helped to alleviate the residual stress;

3) The texture of the highly (220)-oriented Cu films did not change during annealing, but recrystallisation behavior occurred. Both texture and microstructure of the highly (111)-oriented Cu films did not change during the annealing;

4) Annealing relieved residual stresses, and highly (220)-oriented and (111)-oriented nanotwinned Cu films exhibited lower residual stresses after annealing.

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